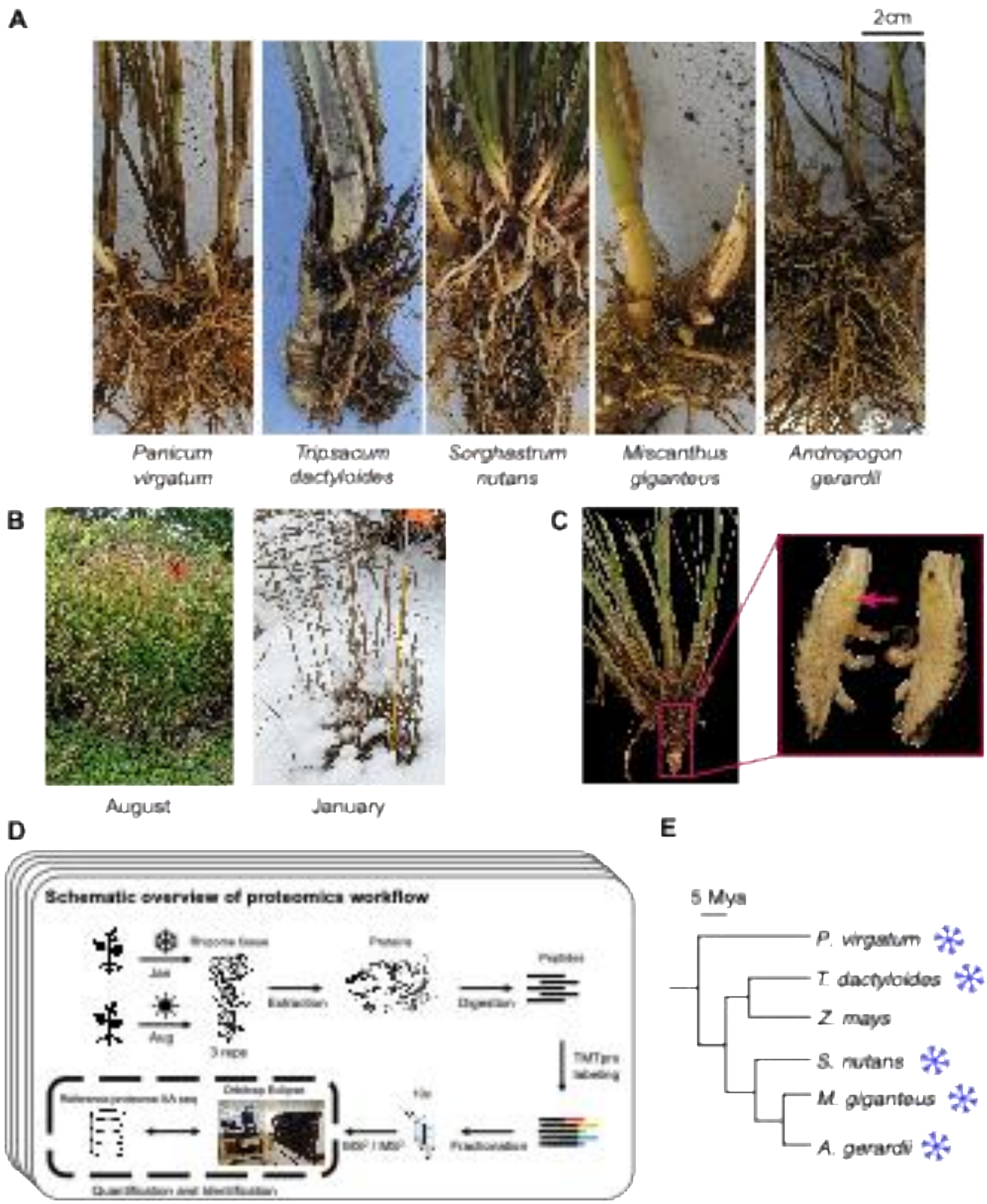


Figure 1. Overview of sampling and proteomics workflow. (A) Rhizomes from the five species sampled in the study: *Tripsacum spp.*, *Panicum virgatum*, *Andropogon gerardii*, *Miscanthus giganteus*, and *Sorghastrum nutans*. **(B)** Field conditions during the two sampling seasons, illustrating the active growth in August and dormancy in January. **(C)** Example of the rhizome tissue sampled for proteomics, shown here for *Tripsacum*. The red arrow highlights the specific region collected. **(D)** Schematic overview of the proteomics workflow. Rhizome samples were collected each season, followed by protein extraction, digestion, and TMTpro labeling. Peptides were fractionated and analyzed via RTS-SPS-MS^3 using an Orbitrap Eclipse mass spectrometer. Each species' reference proteome was used for peptide and protein assignment, enabling quantification and identification. **(E)** Phylogeny of species presented along with *Z. mays* for reference; snowflakes indicate freezing tolerance.

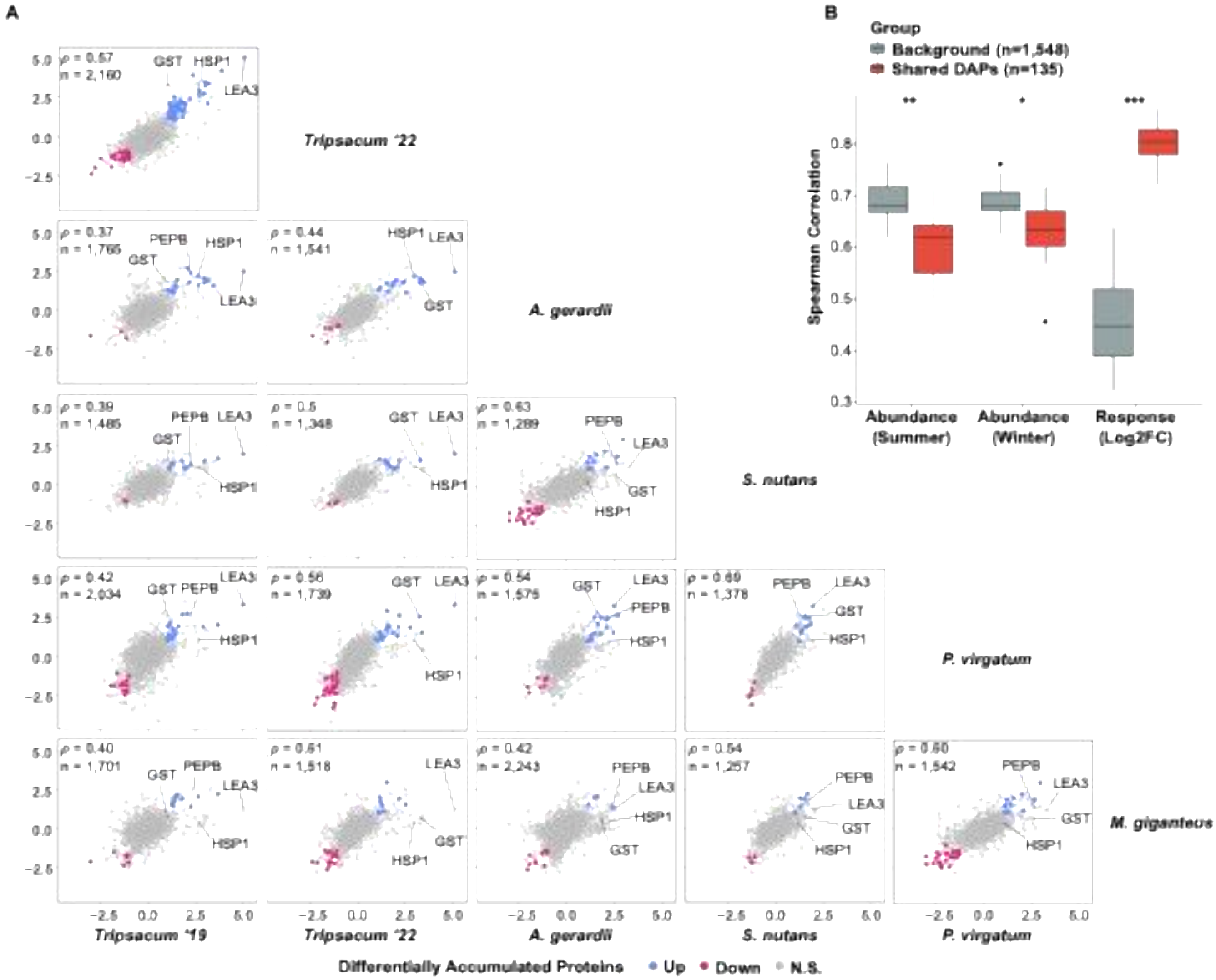
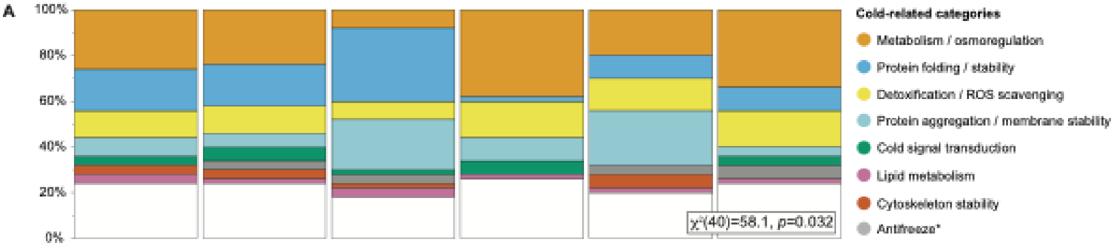


Figure 2. Conservation of seasonal protein accumulation across independently cold-adapted PACMAD grasses. **(A)** Pairwise comparisons of $\log_2$ fold-change (Winter/Summer) in differentially accumulated orthogroups across species. Scatterplots display the correlation of orthogroup-level $\log_2$FC values between species, with shared DAPs elevated in winter shown in blue, reduced in red, and non-significant proteins in gray. Each panel represents a comparison between two species, with the number of shared orthogroups (*n*) and Spearman's correlation coefficient (ρ) displayed. **(B)** Pairwise Spearman correlations for orthogroups quantified in at least four of five species, comparing shared DAPs (red; differentially abundant in ≥2 species) versus background orthogroups (gray; differentially abundant in ≤1 species). Correlations are shown for summer abundance, winter abundance, and seasonal response magnitude ($\log_2$FC winter/summer). Wilcoxon rank-sum test: *P < 0.05, **P < 0.01, ***P < 0.001.

A



B

Rank	*Tripsacum* 2019	*Tripsacum* 2022	*A. gerardii*	*M. giganteus*	*P. virgatum*	*S. nutans*
1	LEA3	LEA3	PEBP	NDUFS1	Embryonic DC-8	Barwin
2	HSP22.0	Abscisic stress ripening	Chitinase	Embryonic DC-8	LEA3	Zeamatin
3	Abscisic stress ripening	Uncharacterized	Actin-depolymerizing 2	Malate synthase	Isoflavone hydroxylase	LEA3
4	TLC	HSP, chloroplastic	Subtilisin-chymotrypsin	Aldose reductase	PEBP	Chitinase
5	Aldose reductase	HSP22.0	LEA-D34	Uncharacterized	Dehydrin	Uncharacterized
6	Dehydrin	HSP I	LEA3	UDP-glucose epimerase	LEA-D34	Chitinase
7	SBP65	WIBG Mago-binding	LEA3	LEA3	LEA3	LTI65
8	PG inhibitor	Chitinase	Uncharacterized	TIF3 subunit G	NDUFS1	UDP-glucose epimerase
9	HSP, chloroplastic	EFHD	Uncharacterized	HSP70 BIP2	LEA14-A	Aldose reductase
10	Dehydrin	HSP16.9A	HSP I	Beta-amylase	Actin-depolymerizing 3	ATP-PFK 2
11	EFHD	Aldose reductase	HSP I	Aldose reductase	GEM-like protein 5	Protein IN2-1
12	GST	HSP70	LEA3	LEA-D34	GST	OEP162
13	Methionine γ-lyase	OEP162	GH18	Reticulon-like	Nematode HSPRO2	Tryptophan synthase
14	OEP162	Dehydrin	HSP I	GEM-like protein 5	Uncharacterized	EFHD
15	DOL monooxygenase	TIL1	HSP I	Ureidoglycolate hydr	Xylanase inhibitor	GST
16	Dehydrin	Chitinase	LTI65	Tryptophan synthase	LTI65	GEM-like protein 5
17	TIL1	Cylicin-1	Ribonuclease 1	PEBP	Actin-depolymerizing 3	HSA32
18	UDP-glucose epimerase	TSJT1	HSP I	Raffinose synthase	LTI65	Anthranilate PRTase
19	HSP16.9A	Uncharacterized	Embryonic DC-8	LTI65	GEM-like protein 5	TVP15
20	1,3-Beta-glucosidase	Aconitase 1	LTI65	Pathogenesis-related 1	Chitinase	Ca-transporting ATPase
21	sHSP17.8	HSP I	HSP, mitochondrial	LEA14-A	GST	Anthranilate PRTase
22	Nitrite reductase	Starch synthase	HSP23.2	PV42a	Embryonic DC-8	Phosphoglycerate mut
23	GST	Malate synthase	LTI65	Tryptophan synthase	OPR1	AAA+ ATPase
24	Transmembrane protein	Heme-binding protein	rRNA N-glycosylase	Peroxidase	UMP-CMP kinase	HSP70 8
25	PEBP	sHSP17.8	EFHD	Tryptophan synthase	SBE II-1	PEBP
26	Aconitase 1	PEP carboxykinase 2	Phosphoglycerate mut	GLO3	HSP70	Reticulon-like
27	KOW	GEM-like protein 5	PEBP	Senescence domain	Pre-mRNA-processing	Phosphoglycerate mut
28	Malic enzyme	Protein DETOX	sHSP26	TSJT1	Embryonic DC-8	DUF255
29	HSP I	UspA	Uncharacterized	GST	GST	Transketolase
30	Actin-depolymerizing 3	sHSP17.8	Pyruvate decarboxylase	Aminotransferase III	Phosphoglycerate mut	AGPase
31	Fe2OG dioxygenase	Cylicin-1	protein-di-S reductase	14-3-3 protein 7	Dehydrin	HSP I
32	Protein DETOX	UspA	Ricin	Ricin B-like lectin	HSP III	Alcohol dehydrogenase
33	sHSP17.8	cpRBP	Peroxidase	DUF255	Aldose reductase	PDH-E1α
34	PEP carboxykinase 2	Fe2OG dioxygenase	Dirigent protein	Anthranilate synthase	OEP162	DJ-1/PfpI
35	UspA	Sucrose synthase	HSP70	AAA+ ATPase	FAD-binding PCMH	HSP II
36	HSP70	Starch synthase	HSP83	DJ-1 homolog B	HSP70	UspA
37	Starch synthase	Beta-glucosidase 31	LEA-D34	OEP162	LEA-D34	GST
38	GST	Dipeptidylpeptidase IV	LEA-D34	Thiamine thiazole	MORF2	Cystatin-1
39	sHSP17.8	Malic enzyme	Dirigent protein	GPT2	HSP, mitochondrial	Cystatin-1
40	GEM-like protein 5	GST	Thioredoxin	Ca-transporting ATPase	Cylicin-1	Dipeptidylpeptidase IV
41	Phloem-specific lectin	Peroxidase	Peroxidase	Peptidase M48	VOC	TATA-binding protein
42	Cylicin-1	Starch synthase	Aconitase 1	Pectinesterase inhibitor	HSP101	AGPase
43	GNAT family	AGPase	Malic enzyme	Pre-mRNA-processing	OEP162	ClpB2, chloroplastic
44	Cytochrome P450	RAB5	Zeamatin	14-3-3 protein 7	ATP-PFK 2	GST
45	Starch synthase	LEA-D34	HSP23.2	DJ-1 homolog D	AAA+ ATPase	CYFBP
46	Starch synthase	Abscisic stress ripening	Aldose reductase	Anthranilate PRTase	SBE II-1	Tryptophan synthase
47	W2	HSP70 8	sHSP26	Galactokinase	Malic enzyme	Zinc-regulated GTPase
48	HSP83	Acyl transferase 5	Embryonic DC-8	GST	Phosphoglycerate K	Dienelactone hydrolase
49	Adenylate kinase	Cytochrome P450	HSP70	PDH-E1α	Chitinase	Malic enzyme
50	IPS3	GST-N	sHSP	GLO3	PCYOX	Peroxidase

Figure 4. Overview of cold-related functional categories represented among the top 50 most highly upregulated rhizome proteins across five PACMAD grasses. (A) Mosaic plot of cold-related functional categories across species. This plot visualizes the distribution of proteins among cold-related functional categories for each species. The relative abundance of each category within a species is shown, highlighting distinct functional profiles and species-specific variations in cold tolerance mechanisms. A Chi-square test for independence ($\chi^2[40] = 58.1$, $p = 0.032$) was performed to evaluate variation across species. **(B)** Visual representation of cold-related functional categories in rhizome proteins of five Andropogoneae species. This figure illustrates the top 50 most abundant proteins within the rhizomes of five Andropogoneae species. Proteins are classified into functional categories relevant to cold-related mechanisms, such as metabolism/osmoregulation, cold signal transduction, membrane stability, lipid metabolism, and others. Color-coding facilitates a comparative visualization of cold tolerance mechanisms across species. Antifreeze proteins are marked by an asterisk, as their classification is based solely on protein homology.

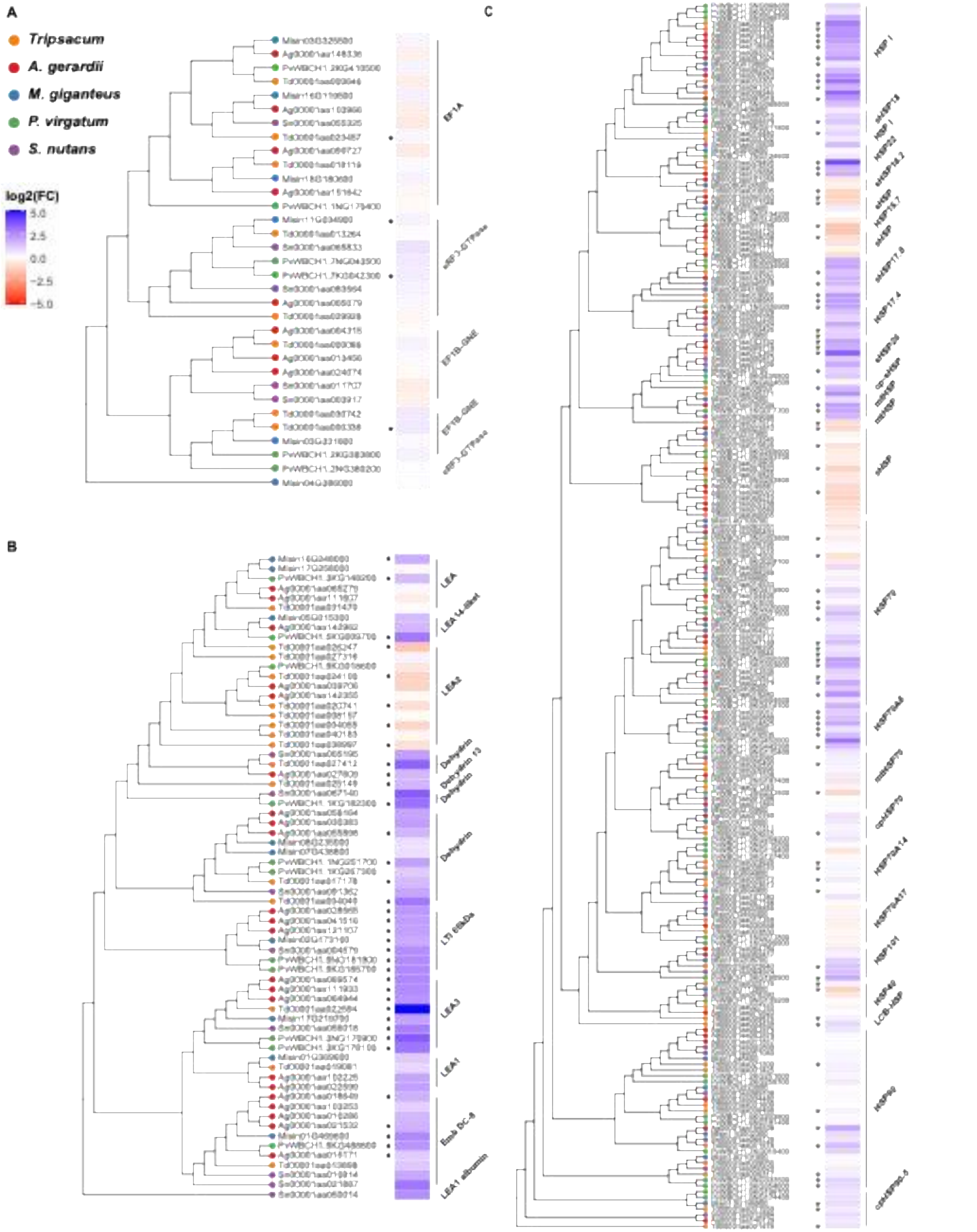


Figure 5. Phylogenetic analysis and seasonal abundance dynamics of selected protein families across five PACMAD grasses (*Tripsacum spp.*, *P. virgatum*, *A. gerardii*, *M. giganteus*, and *S. nutans*). Protein identifiers represent species-specific sequences, and brackets denote functional group annotations, merging adjacent orthogroups with similar functions for clarity. Heat maps illustrate $\log_2$ fold-change ($\log_2$FC) in protein abundance between winter and summer conditions, where blue indicates higher abundance in winter, and red signifies higher summer abundance. Panels show distinct protein families: **(A)** translation elongation factor 1 proteins (EF1; control group) , **(B)** late embryogenesis abundant (LEA) proteins, and **(C)** heat shock proteins (HSPs). Asterisks (*) mark proteins exhibiting statistically significant differential accumulation ($p < 0.05$).

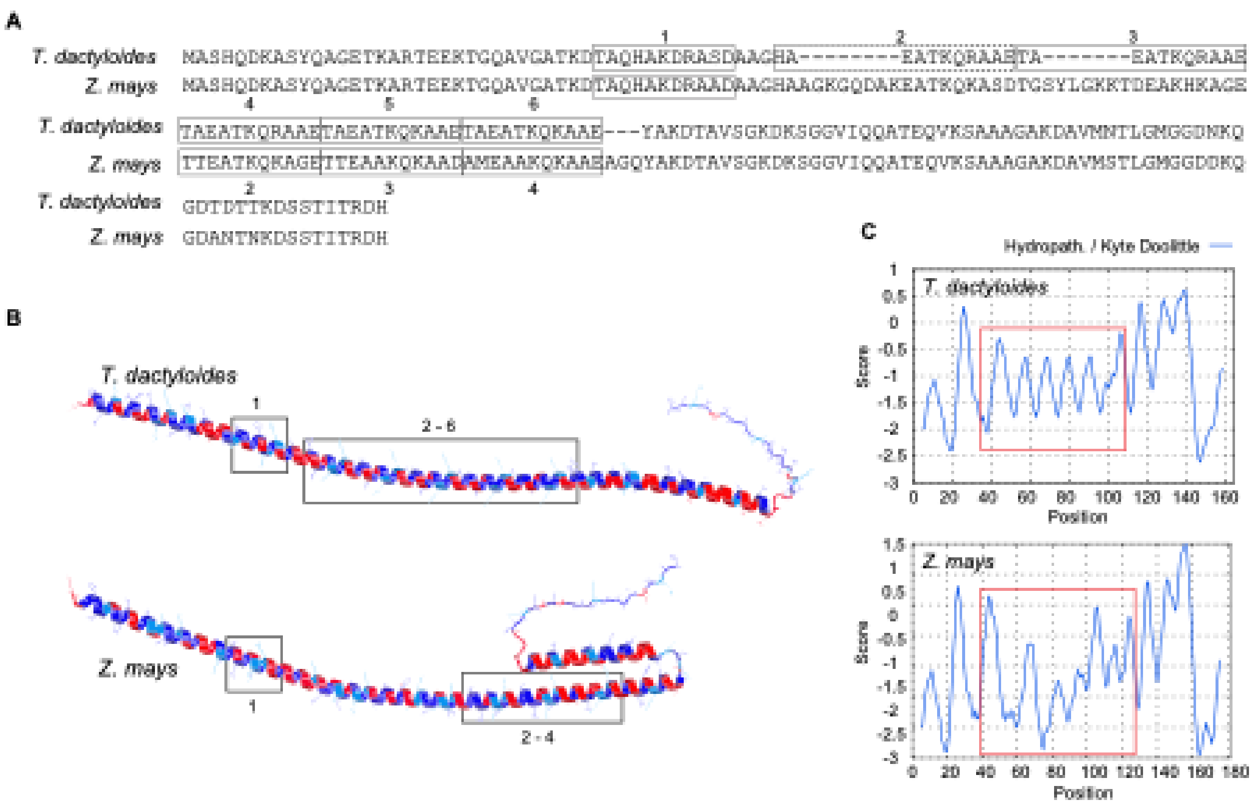


Figure 6. **LEA3 protein sequence alignment, structure and hydrophobicity. (A**) The alignment shows the LEA3 proteins from *T. dactyloides* (Td00001aa022594_T002) and, *Z. mays* (Zm00001eb294480), with the repeating 11-mer LEA3 motifs highlighted and labeled numerically. The dashed motif deviates from Dure's LEA3 motifs in the first position. (**B**) Predicted secondary structure of LEA3 proteins using AlphaFold3, showing the conserved alpha-helical domains across *T. dactyloides* (top) and *Z. mays* (bottom). LEA motif regions corresponding to labeled sequence motifs (e.g., 1, 2–6) are boxed. Red and blue indicate regions of high hydrophilicity and hydrophobicity, respectively. **(C)** Kyte-Doolittle hydropathy plots for LEA3 proteins. The X-axis represents residue position, while the Y-axis indicates hydropathicity scores (positive values for hydrophobic regions, negative for hydrophilic regions). 11-mer motif regions are highlighted with red boxes, corresponding to key domains in (B)**.**